



$p > 0 \Rightarrow$ compression

$$\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = -p \Rightarrow \sigma_{ii} = -3p$$

$$e_{ii} = \frac{\sigma_{ii}}{(3\lambda + 2\mu)} = \frac{-3p}{3\lambda + 2\mu} = \frac{-p}{\lambda + \frac{2}{3}\mu} = \frac{-p}{K}$$

$$e_{ii} = \frac{\Delta V}{V} \Rightarrow \frac{\Delta V}{V} = \frac{-p}{K}$$

$$\frac{\Delta V}{V} = \frac{-p}{K} = \frac{-p}{\frac{E}{3(1-2\nu)}} = -\frac{3(1-2\nu)}{E} p \Rightarrow \Delta V = -\frac{3(1-2\nu)pV}{E}$$

$$\Delta V \leq 0 \text{ quando } p \geq 0 \Rightarrow \overbrace{-3(1-2\nu)}^{\Delta V} \cancel{p} \cancel{V} \leq 0$$

$p \neq 0$
 $V \neq 0$
 $E \neq 0$

$$(1-2\nu) \geq 0 \Rightarrow$$

$$\boxed{\nu \leq \frac{1}{2}}$$